Reza Mohammadvand

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reza-mohammadvand reza-mohammadvand

ABOUT ME

I am a recent MSc graduate in Computer Networks with a strong interest in pursuing a PhD to advance my academic and research career. My primary goal is to explore cuttingedge research at the intersection of network infrastructure, cloud computing, and edge intelligence. I am particularly interested in addressing challenges related to low-latency and high-reliability communication systems, distributed intelligence, and scalable architectures for next-generation technologies such as 5G/6G, IoT, and digital twin systems.

My objective is to join a research-oriented PhD program where I can deepen my expertise, contribute to innovative projects, and collaborate with experts in the field. I aim to develop practical solutions grounded in rigorous theoretical work, with the long-term vision of becoming a researcher and academic who drives technological innovation in intelligent and connected systems.

EDUCATION

Master of Science in Computer Engineering 2024
Iran University of Science and Technology, Tehran, Iran

- ✓ *GPA*: 17.66/20
- ✓ *Related coursework*: Advanced computer networks, advanced network security, wireless and mobile networks, wireless communications, distributed systems
- ✓ *Thesis:* Thesis: Using machine learning algorithms in Edge intelligence for Ultra-Reliable and Low-Latency Communications in the industrial metaverse

Bachelor of Science in Computer Engineering 2022
Islamic Azad University, Dezful, Iran

- ✓ *GPA*: 17.75/20
- ✓ *Related coursework*: Digital Logic Circuits, Data Structures, Computer Architecture, Theory of Languages and Automata, Databases, Operating Systems, Computer Networks

RESEARCH PROJECTS

Edge Intelligence-based URLLC for Digital Twin-enabled Metaverse

- ✓ Designed and implemented a novel edge intelligence-based framework to support ultra-reliable and low-latency communication (URLLC) in a Digital Twin-enabled Metaverse environment
- ✓ Utilized reinforcement learning and edge computing to reduce both latency and energy consumption in real-time industrial applications
- ✓ Achieved notable improvements in end-to-end delay and energy efficiency under dynamic network conditions

PRESENTATIONS

- ✓ Oral Presentation: "Using Machine Learning Algorithms in Edge Intelligence for Ultra-Reliable and Low-Latency Communications in the Industrial Metaverse", 29th International Computer Conference, Computer Society of Iran (CSICC) (6 Feb 2025)

SKILLS

Programming and AI

- ✓ Designed and tested **Machine Learning** models using **Python**.
- ✓ Applied automation and scripting in system administration and data analysis tasks.

Networking and Systems

- ✓ Configured and troubleshooted enterprise networks using **CCNA** and **MCSA** knowledge.
- ✓ Deployed and maintained systems in **Linux** environments (**LPIC-1 & LPIC-2**).

Cloud and Virtualization

- ✓ Explored core **cloud computing** concepts and implemented basic virtualization tasks.

Languages

- Persian: Native language
- English: Advanced level (academic and professional proficiency)
  - ✓ *Writing:* Demonstrated through the writing of my master's thesis and research papers in English.
  - ✓ *Reading:* Regularly read academic papers, technical documentation, and research articles related to networking and cloud computing.
  - ✓ *Speaking and Listening:* Practiced through academic discussions, online meetings, and English-speaking study groups.

EXPERIENCE

IT Intern — Jul 2021 – Sep 2021

Dez Dam Power Plant, Khuzestan, Iran

- ✓ Completed a structured internship in the IT department of one of Iran's largest hydroelectric power plants.
- ✓ Assisted with network monitoring, hardware troubleshooting, and data backup management.
- ✓ Gained practical experience with enterprise IT infrastructure and support.

IT Specialist — Aug 2025 – Present

Mehr Investment Company, Tehran, Iran

- ✓ Providing technical support for hardware, software, and network-related issues in a financial enterprise environment.
- ✓ Performing routine system maintenance, updates, and security checks to ensure business continuity.
- ✓ Assisting with user account management, peripheral device setup, and troubleshooting Windows-based systems.

PUBLICATIONS

Journal Articles

- ✓ Mohammadvand, R., & Mozayani, N. (2025). Using Machine Learning Algorithms in Edge Intelligence for Ultra-Reliable and Low-Latency Communications in the Industrial Metaverse. *2025 29th International Computer Conference, Computer Society of Iran (CSICC)*, pp. 1–5. DOI: **10.1109/CSICC65765.2025.10967422**.

- ✓ Mohammadvand, Reza, Nasser Mozayani, and Saeed Khoshkholghi. "An edge intelligence framework with reinforcement learning for digital twins in industrial metaverse." Cluster Computing 29, no. 9 (2026): 541. DOI: **10.1007/s10586-026-06380-7.**